MULTIVARIABLE CALCULUS • UNIT 3.01

C Multivariable Optimization, Hessians & Lagrange Multipliers

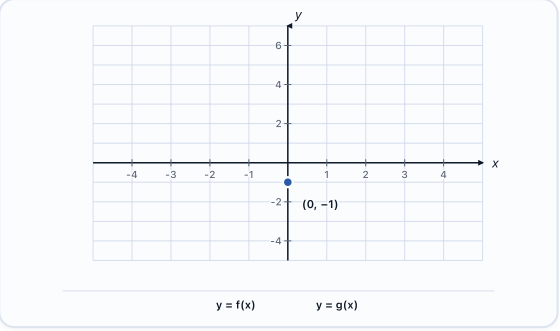
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Multivariable Optimization, Hessians & Lagrange Multipliers Practice Exercises

1 MCQ

Consider the function $z = f(x, y) = x^2 + y^2$. Let $x = 2t$ and $y = 3t$. Use the Multivariable Chain Rule to find $\frac{dz}{dt}$ when $t = 1$.



A 10

B 13

C 26

D 52

2 MCQ

Evaluate the double integral of the function $f(x, y) = 2xy$ over the rectangular region $R = [0, 2] \times [0, 3]$ using Fubini's Theorem. Set up the iterated integral by integrating with respect to y first, then x.

A 12

B 18

C 6

D 24

3 MCQ

For $f(x, y) = 7x^2 + 4y^2$, find $f_y(-3, 1)$.

A -24

B 4

C 8

D -42

4 MCQ

Consider the function $f(x, y) = x^2 + y^2$. Find the value of $f_{xx}(2, 3)$.

A 2

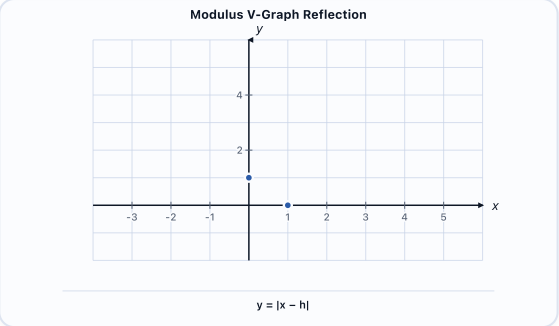
B 4

C 6

D 0

5 MCQ

Let the linear scalar potential be $h(u, y) = 15u + 8y - 16$. Compute the squared gradient norm $\|\nabla h\|^2$.



A 289

B 293

C 281

D 304

6 MCQ

Consider the surface S which is the portion of the plane $z = 2x + 3y + 1$ lying above the rectangular region $R = [0, 1] \times [0, 2]$ in the xy-plane. Calculate the surface area of S.

A $\sqrt{14}$

B $2\sqrt{14}$

C $3\sqrt{14}$

D $4\sqrt{14}$

7 MCQ

Consider the vector field $\mathbf{u}(s, y, z) = \langle 22s,\ 7y,\ 7z \rangle$ on R^3. Compute the scalar divergence $\nabla \cdot \mathbf{u}$.

A 58

B 40

C 29

D 36

8 MCQ

Consider the vector field $\mathbf{F}(x, y) = \langle 3x, 2y \rangle$. Evaluate the vector field at the point $(2, 4)$.

A $\langle 6, 8 \rangle$

B $\langle 8, 6 \rangle$

C $\langle 5, 6 \rangle$

D $\langle 6, 5 \rangle$

9 MCQ

Consider the function $f(x, y) = x^2 + y^2 - 4x - 6y + 13$. Find the coordinates of the critical point for this function.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	13	10	8	5

A (-2, -3)

B (2, 3)

C (4, 6)

D (0, 0)

10 MCQ

Consider the vector field $\mathbf{h}(u, y, z) = \langle 4u,\ 10y,\ 5z \rangle$ on R^3. Compute the scalar divergence $\nabla \cdot \mathbf{h}$.

A 28

B 9

C 23

D 19

11 MCQ

Consider the function $f(x, y) = x^3 + y^3 - 3axy$, where a is a positive constant. It is known that $f(x, y)$ has a critical point at (x_0, y_0) where $x_0 = 2$ and $y_0 = 2$. Find the value of a.

A 1

B 3

C 2

D 4

12 MCQ

Consider the vector field $\mathbf{f}(x, y, z) = \langle 12x,\ 20y,\ 11z \rangle$ on R^3. Determine the scalar divergence $\nabla \cdot \mathbf{f}$.

A 23

B 43

C 47

D 55

13 MCQ

Let the multivariable function u be defined by $u(s, y) = 14s^2y + 24y^2$. Compute the partial derivative $\frac{\partial u}{\partial s}(6, 10)$.

A 1680

B 1656

C 1684

D 1694

14 MCQ

Calculate $\nabla f(-2,-1)$ for $f(x,y) = 4x^2 + 8y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	4	8	12

A (-8, -8)

B (-16, -16)

C (16, 16)

D (-16, -14)

15 MCQ

Find $\nabla f(1,3)$ for $f(x,y) = 7x^2 + 9y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	7	9	16

A (54, 14)

B (7, 27)

C (-14, -54)

D (14, 54)

16 MCQ

Establish $\nabla f(-3,-6)$ for $f(x,y) = 6x^2 + 4y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	6	4	10

A (-48, -36)

B (-18, -24)

C (-36, -48)

D (36, 48)

17 MCQ

Find $\nabla f(-1,4)$ for $f(x,y) = x^2 + 2y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	1	2	3

A (16, -2)

B (-2, 16)

C (-1, 8)

D (2, -16)

18 MCQ

Establish $\nabla f(-1, 1)$ for $f(x, y) = 3x^2 + 9y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	3	9	12

A (18, -6)

B (-6, 18)

C (-3, 9)

D (6, -18)

19 MCQ

Find $\nabla f(-1, -2)$ for $f(x, y) = 4x^2 + y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	4	1	5

A (-8, -4)

B (-4, -8)

C (-4, -2)

D (8, 4)

20 MCQ

Find $\nabla f(-2, 2)$ for $f(x, y) = 4x^2y + 4xy^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	0	0	8

A (16, 16)

B (-16, -16)

C (-15, -16)

D (-32, -32)

21 MCQ

Deduce $\nabla f(4, -2)$ for $f(x, y) = 4x^2y + 5xy^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	0	0	9

A (-64, -80)

B (-16, -44)

C (-44, -16)

D (44, 16)

22 MCQ

Compute $\nabla f(1, 3)$ for $f(x, y) = 4x^2y + 5xy^2$.

A (24, 30)

B (34, 69)

C (-69, -34)

D (69, 34)

23 MCQ

Obtain $\nabla f(4, 3)$ for $f(x, y) = 5x^2y + 3xy^2$.

A (120, 72)

B (152, 147)

C (-147, -152)

D (147, 152)

24 MCQ

Determine $\nabla f(-3, -1)$ for $f(x, y) = 5x^2y + 2xy^2$.

A (30, 12)

B (57, 32)

C (32, 57)

D (-32, -57)

25 MCQ

Deduce $\nabla f(2, -2)$ for $f(x, y) = x^2y + 5xy^2$.

A (-8, -40)

B (-36, 12)

C (12, -36)

D (-12, 36)